

Supplementary contact-recovery summary

Compact recovery heatmaps for whole local environments, basic residues, and donor atom-phosphate oxygen contacts.

(a)

5 A crystal environment contact recovery

All non-water heavy-atom contacts around the phosphosite marker atoms.

Case	Crystal contacts	PhosphoFill top-1	PhosphoFill best-of-3	PyTMs default	PyTMs optimised	PTM-Psi
CDK2 TPO 160 (2CCH)	8	8/8 (100%)	8/8 (100%)	6/8 (75%)	5/8 (62%)	5/8 (62%)
CDK2 TPO 160 (4EOJ)	9	9/9 (100%)	9/9 (100%)	7/9 (78%)	7/9 (78%)	5/9 (56%)
ERK2 TPO 185 (4IZA)	9	8/9 (89%)	8/9 (89%)	6/9 (67%)	6/9 (67%)	5/9 (56%)
ERK2 TPO 185 (5V62)	7	7/7 (100%)	7/7 (100%)	4/7 (57%)	7/7 (100%)	4/7 (57%)
ERK2 PTR 187 (4IZA)	6	6/6 (100%)	6/6 (100%)	6/6 (100%)	3/6 (50%)	5/6 (83%)
ERK2 PTR 187 (5V62)	4	4/4 (100%)	3/4 (75%)	2/4 (50%)	4/4 (100%)	0/4 (0%)

(b)

4 A basic-residue contact recovery

Recovery of crystal Arg/Lys/His residues contacting phosphate oxygens.

Case	Crystal basic	PhosphoFill top-1	PhosphoFill best-of-3	PyTMs default	PyTMs optimised	PTM-Psi
CDK2 TPO 160 (2CCH)	3	3/3 (100%)	3/3 (100%)	2/3 (67%)	1/3 (33%)	3/3 (100%)
CDK2 TPO 160 (4EOJ)	3	3/3 (100%)	3/3 (100%)	2/3 (67%)	2/3 (67%)	3/3 (100%)
ERK2 TPO 185 (4IZA)	2	2/2 (100%)	2/2 (100%)	2/2 (100%)	1/2 (50%)	2/2 (100%)
ERK2 TPO 185 (5V62)	3	3/3 (100%)	3/3 (100%)	2/3 (67%)	3/3 (100%)	2/3 (67%)
ERK2 PTR 187 (4IZA)	2	2/2 (100%)	2/2 (100%)	2/2 (100%)	1/2 (50%)	0/2 (0%)
ERK2 PTR 187 (5V62)	2	2/2 (100%)	2/2 (100%)	0/2 (0%)	2/2 (100%)	0/2 (0%)

(c)

4 A donor atom-phosphate oxygen recovery

Recovery of donor atom contacts to any phosphate oxygen.

Case	Crystal donors	PhosphoFill top-1	PhosphoFill best-of-3	PyTMs default	PyTMs optimised	PTM-Psi
CDK2 TPO 160 (2CCH)	6	6/6 (100%)	6/6 (100%)	3/6 (50%)	1/6 (17%)	4/6 (67%)
CDK2 TPO 160 (4EOJ)	6	6/6 (100%)	6/6 (100%)	4/6 (67%)	3/6 (50%)	4/6 (67%)
ERK2 TPO 185 (4IZA)	3	3/3 (100%)	3/3 (100%)	3/3 (100%)	1/3 (33%)	3/3 (100%)
ERK2 TPO 185 (5V62)	5	5/5 (100%)	5/5 (100%)	4/5 (80%)	5/5 (100%)	3/5 (60%)
ERK2 PTR 187 (4IZA)	3	3/3 (100%)	3/3 (100%)	3/3 (100%)	1/3 (33%)	0/3 (0%)
ERK2 PTR 187 (5V62)	4	4/4 (100%)	4/4 (100%)	0/4 (0%)	4/4 (100%)	0/4 (0%)

Cell color: >=75% recovered 40-74% recovered <40% recovered

All cells report recovered crystal contacts / total crystal contacts.

Detailed residue and donor-atom labels are provided in figs3_contact_recovery_detail.tsv.